Michael Sidoti

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EDUCATION

Cornell University **Ithaca, NY**
Bachelor of Arts in Computer Science and Mathematics | GPA: 3.85 *Expected Graduation: May 2028*
Relevant Coursework: Data Structures and Algorithms, Object-oriented Programming, Analysis of Algorithms, Machine Learning (Graduate level), Data Structures and Functional Programming, Computer System Organization and Programming, Probability Theory, Combinatorics, Linear Algebra, Multivariable Calculus, Differential Equations

EXPERIENCE

Amazon Web Services **May 2026 – Aug. 2026**
Software Development Engineer Intern *Herndon, VA*
- Designed an event-driven AWS notification system for an internal service used by 10,000 engineers and 5,000 resolver groups, using DynamoDB, EventBridge, Lambda, SES, SQS DLQs, and CloudWatch alarms to alert teams of routing, paging, and permissions changes.
- Modeled scalable DynamoDB tables for notification events and user preferences using resolver group partition keys, timestamp-based sort keys, TTL expiry, and channel-specific configuration for batched notification delivery.
- Implemented notification preference APIs for setting, retrieving, and removing resolver group subscriptions, enabling teams to manage event types, delivery channels, and notification preferences.
- Designed fault-tolerant dispatch logic with persisted event state, per-channel statuses, retry handling, dead-letter queues, and extensible SES/SNS/Slack channel formatting.

Undergraduate Research Assistant – Math + AI Lab **Jan. 2026 – Present**
Cornell University *Ithaca, NY*
- Selected from a competitive pool of applicants to conduct research in Cornell's largest fully funded AI research lab.
- Conducted AI research under Prof. Alex Townsend, translating Chebfun, a mathematically sophisticated MATLAB numerical computing library, into Python while preserving numerical semantics, API behavior, and test-driven correctness.
- Helped evaluate LLM coding agents on large-scale scientific software translation, documenting limitations around overloaded syntax, floating-point conventions, object dispatch, and numerical edge cases.
- Presented team research to 40+ students and faculty across different Math + AI Lab research groups and co-authored "Translating Chebfun from MATLAB to Python," a research paper planned for arXiv submission.

Operations Research & Information Engineering Project Team **Dec. 2025 – May 2026**
Cornell University *Ithaca, NY*
- Built a Python simulator for single-server queueing systems with discrete-rank scheduling policies, tracking arrivals, service progress, completions, and mean job delay.
- Developed a bisection-based simulation algorithm to skip unnecessary rank-breakpoint events versus naive discrete-event simulation.
- Optimized simulator performance with NumPy vectorization, achieving a 108x speedup over naive discrete-event simulation on 5,000-job, high-resolution benchmarks.
- Collaborated with a PhD mentor and ORIE faculty through twice-weekly meetings on queueing theory, scheduling policies, and simulation optimization.

Teaching Assistant – CS 5820: Analysis of Algorithms, CS 2800: Discrete Math **Sep. 2025 – Present**
Cornell University *Ithaca, NY*
- Led weekly 2-hour discussion sections for 30 students on algorithm design, graph theory, proof techniques, asymptotic analysis, and probability theory.
- Held tutoring sessions for 10 hours weekly, supporting 300+ students across Analysis of Algorithms and Discrete Mathematics coursework.
- Graded homework assignments and contributed to exam and quiz questions covering asymptotic analysis, recurrence relations, induction, graph theory, and probability theory.

Artificial Intelligence Trainer **Jan. 2025 – Present**
DataAnnotation *Remote*
- Passed selective technical screening to train AI models in computer science, mathematics, and physics.
- Evaluated LLM outputs through reinforcement learning from human feedback workflows, reviewing complex computer science, mathematics, and physics reasoning tasks to improve model accuracy.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, Go, OCaml
Frameworks: FastAPI, JUnit, Spring Boot, React, Tailwind, Node.js
Libraries: pandas, NumPy, Matplotlib
Cloud/Tools: AWS, DynamoDB, Lambda, EventBridge, CloudWatch, Git, Docker